UNIVERSITY OF TECHNOLOGY, JAMAICA
SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY
Programming 2: Tutorial/Practical Worksheet #6

Tutorial Questions

1. In your own words explain linear and binary search algorithms.
2. Use the selection sort, insertion sort and bubble sort on the following array:

   9, 20, 7, 18, 8, 41, 35, 6, 3, 13

   (No code required)
3. Given the list of numbers:

   88, 30, 500, 70, 150, 40, 8, 450, 55, 150

   Using insertion sort and the selection sort algorithms (descending order).
   a. What will be the result are three (3) turns?
   b. What will be the result after five (5) turns?

   (No code required)
4. Write a single C statement that will initialize a two-dimensional array of integers with 4 test grades for each of 3 students. Student 1 has the following grades: 75, 65, 45, 20. Student 2 has the following grades: 85, 90, 100, 70. Student 3 has the following grades: 40, 30, 35, 60.
5. Create a 2-dimensional array that accepts from the user 4 test grades each for 3 students. Write code that will
   a. Print the 4 test grades for the first student1.
   b. Print the second test grades for all the students2.
   c. Print all grades for all students.
   d. For each student, determine and print the highest grade obtained of the 4 tests.
   e. Find the average grade for each student.
   f. Find the average grade for each test.

In the Laboratory

Complete any unfinished questions from the tutorial section or **from the previous worksheet**. Implement and execute the following programs using C only:

1. Convert the following sorting pseudocode functions into C, making any necessary modifications for implementation. Define appropriate function prototypes for each. In the

main function, initialize three separate arrays of size 10 and populate them with random integers from 1 to 100 inclusive. Demonstrate the functionality of each algorithm by sorting one of the arrays and printing the results. **Note:** You are not required to memorize these code implementations for any assessment.

```
FUNCTION BubbleSort(Integer Array Arr, Integer SIZE)
   DECLARE temp, i, j AS INTEGER
   FOR i = 0 TO SIZE – 1 STEP 1
      FOR j = 0 TO SIZE – i – 2 STEP 1
         // IF current is greater than next, swap them
         IF Arr[j] > Arr[j + 1] THEN
            temp = Arr[j]
            Arr[j] = Arr[j + 1]
            Arr[j + 1] = temp
         ENDIF
      ENDFOR
   ENDFOR
ENDFUNCTION

FUNCTION SelectionSort(Integer Array Arr, Integer SIZE)
   DECLARE temp, minIndex, I, j AS INTEGER
   FOR i = 0 TO SIZE – 2 STEP 1
      minIndex = i
      // Find the smallest element in the rest of the array
      FOR j = i + 1 TO SIZE – 1 STEP 1
         IF Arr[j] < Arr[minIndex] THEN
            minIndex = j
         ENDIF
      ENDFOR

      // Swap the found minimum element with the first element of the unsorted part
```

```
        IF minIndex != i THEN
            temp = Arr[i]
            Arr[i] = Arr[minIndex]
            Arr[minIndex] = temp
        ENDIF
    ENDFOR
ENDFUNCTION
```

```
FUNCTION InsertionSort(Integer Array Arr, Integer SIZE)
    DECLARE i, j, key AS INTEGER
    FOR i = 1 TO SIZE – 1 DO
        key = Arr[i]
        j = i – 1

        // Shift elements of Array[0..i-1] that are greater than key
        // to one position ahead of their current position
        WHILE j >= 0 AND Arr[j] > key DO
            Arr[j + 1] = Arr[j]
            j = j – 1
        END WHILE

        // Place the key into its correct "hole"
        Arr[j + 1] = key
    ENDFOR
ENDFUNCTION
```

1. A small retail shop records daily sales amounts. Since sales are entered as they happen, the manager wants the list to always remain sorted. Write a C program that stores sales amounts for 7 days. Sorts them in ascending order using **Insertion Sort**. Displays the sorted sales amounts. Calculates the median sales value.

2. A small community library stores book IDs in random order. The librarian wants them sorted numerically before printing labels. Write a C program that accepts 10 book ID numbers. Sorts them in ascending order using **Bubble Sort**. Displays the sorted list. Counts and displays the number of swaps performed.
3. A company wants to review employee salaries from lowest to highest to determine raise eligibility. Write a C program that Stores employee salaries. Sorts them in ascending order using **Selection Sort**. Displays:
    a. The lowest salary
    b. The highest salary
    c. The sorted salary list

Challenge

Use modularity and 2D-subscripted array to solve the following problem. A company has four salespeople (1 to 4) who sell five different products (1 to 5). Once a day, each salesperson passes on a slip for each different type of product sold. Each slip contains:

a. The salesperson number
b. The product number
c. The total dollar value of that product sold that day

Thus, each salesperson passes in between 0 and 5 sales slips per day. ***Assume that the information from all of the slips for last month is available***.

Requirement: Write a program that will read all this information for last month's sales and summarize the total sales by salesperson by product. All totals should be stored in the double-subscripted array sales. After processing all the information for last month, print the results in tabular format with each column representing a particular salesperson and each row representing a particular product. Cross total each row to get the total sales of each product for last month; cross total each column to get the total sales by salesperson for last month. Your tabular printout should include these cross totals to the right of the totaled rows and to the bottom of the totaled columns.